

Deep Learning Tutorial

MNIST Dataset

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Core and Visual Computing Group

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Mixed National Institute of Standards and Technology (MNIST) Dataset

- MNIST Dataset contains:
 - Image files of handwritten digits
 - Files that are 28 x 28 pixels in dimension
 - 70k of images: 10k for testing, 60k for training
- MNIST Dataset is:
 - Commonly used for training
 - Available as a Keras dataset

MNIST Dataset Sample

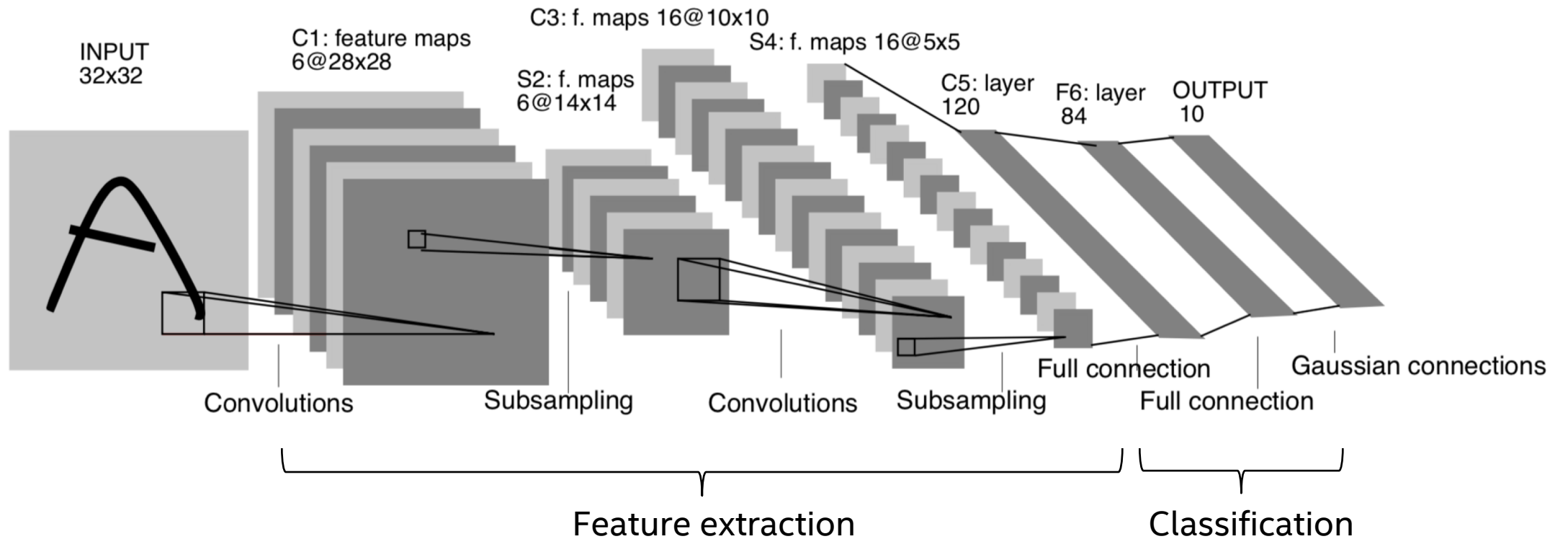


SOURCE: By Josef Steppan - Own work, CC BY-SA 4.0,
<https://commons.wikimedia.org/w/index.php?curid=64810040>

Convolutional Neural Network (CNN or ConvNet)

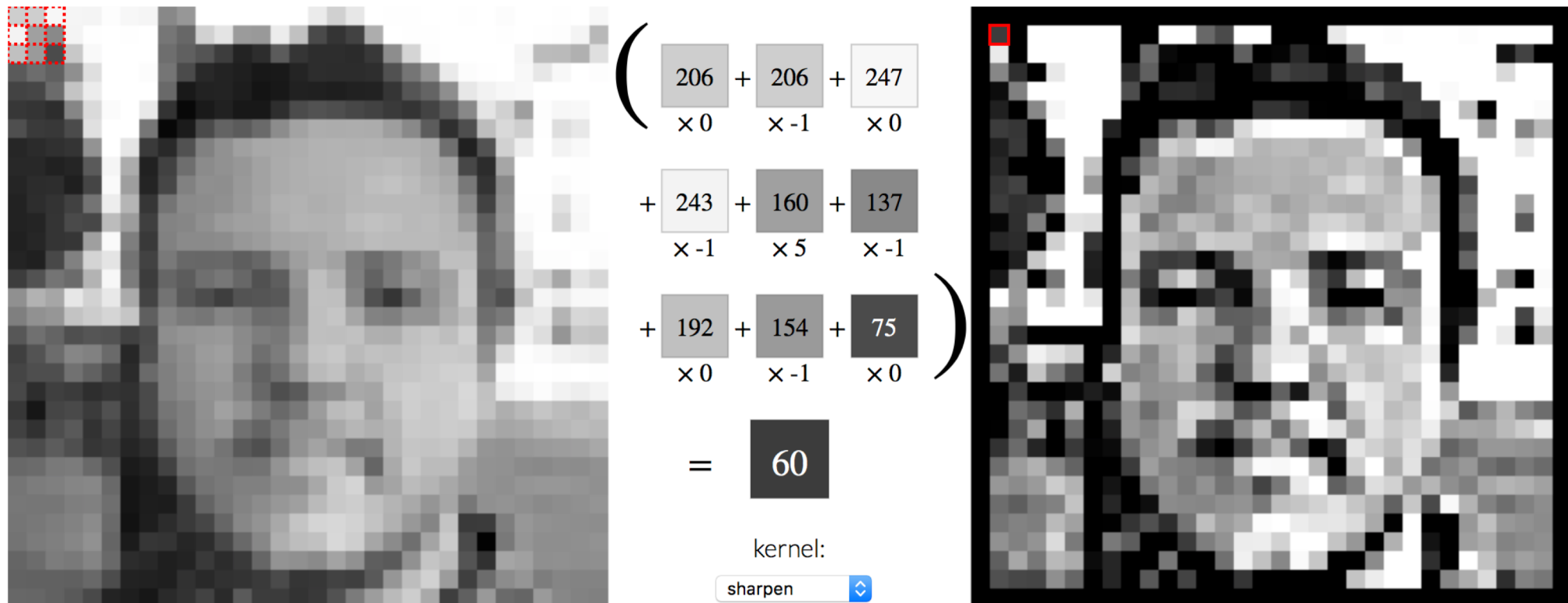
- Feed-forward Artificial Neural Network
- Two major components:
 - Feature detection layers
 - Classification layers
- CNN example – LeNet:
 - LeNet-5, Yann LeCun et al. 1998
 - 7-level CNN

LeNet-5



SOURCE: Gradient Based Learning Applied to Document Recognition - <http://yann.lecun.com/exdb/publis/psgz/lecun-98.ps.gz>

Image Kernels



The diagram illustrates the application of an image kernel. On the left, the "input image" shows a grayscale face. A 3x3 region in the top-left corner is highlighted with a red dashed box. In the center, the kernel calculation is shown as a 3x3 grid of values multiplied by their respective input pixel values, summed to produce the output pixel value of 60. The calculation is as follows:

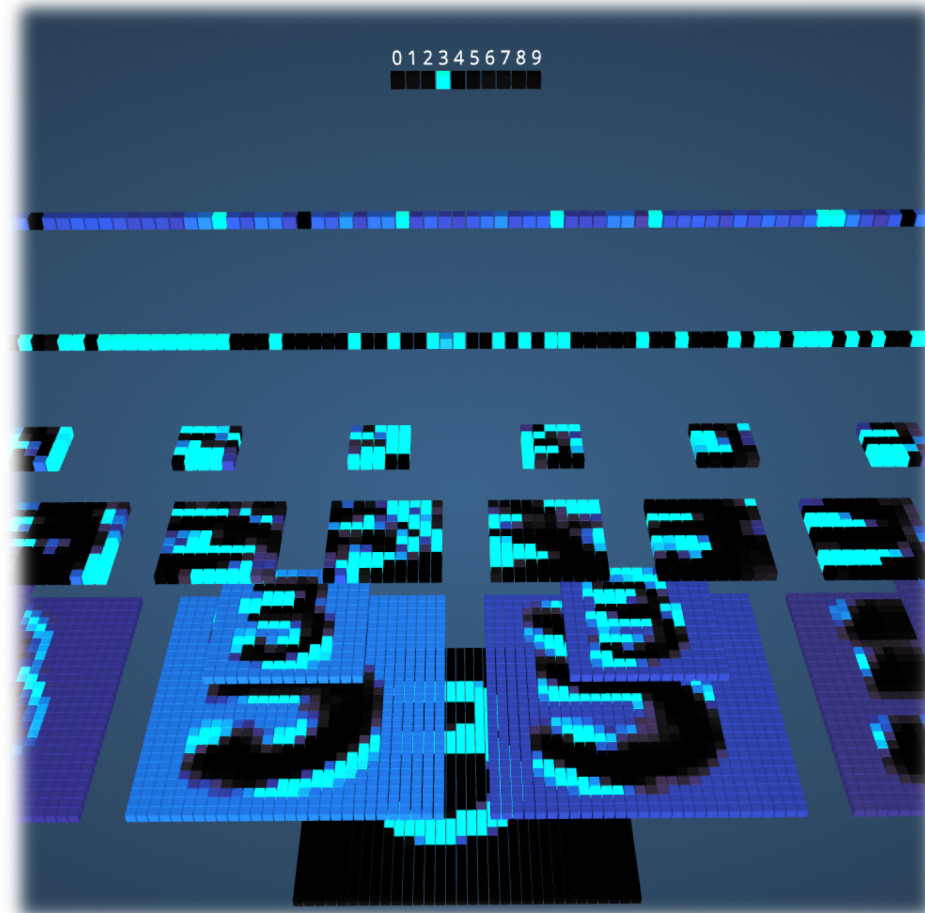
$$\begin{pmatrix} 206 \times 0 & 206 \times -1 & 247 \times 0 \\ + 243 \times -1 & 160 \times 5 & 137 \times -1 \\ + 192 \times 0 & 154 \times -1 & 75 \times 0 \end{pmatrix} = 60$$

Below the calculation, the kernel is identified as "sharpen" with a dropdown arrow.

On the right, the "output image" shows the result of the sharpening operation. A small red square in the top-left corner indicates the output pixel value of 60.

SOURCE: <http://setosa.io/ev/image-kernels/>

3D Interactive Demo



SOURCE: <http://scs.ryerson.ca/~aharley/vis/conv/>

